



Republic of the Philippines
Laguna State Polytechnic University
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College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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I. Objective: Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer:

The objective of this program is to create a simple IT Automation Incident Ticket Manager that can organize and manage incident tickets created by automation bots. The program allows the user to add, display, search, remove, and count incident tickets. Through this activity, I can apply the concepts I have learned about Object-Oriented Programming, especially classes, objects, encapsulation, and methods, while also using a linear data structure to store the incident tickets.

II. Program Operations: Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

Code Line:

```
ticket = IncidentTicket(incident_id, bot, description)
self.tickets.append(ticket)
```

Explanation:

This creates a new IncidentTicket object using the Incident ID, Bot, and Short Description entered by the user. The append() method then adds the new ticket to the tickets list so it can be managed by the program.

Display Incidents:

Code Line:

```
for ticket in self.tickets:
    ticket.display_ticket()
```

Explanation:

The program goes through each ticket stored in the list. It then uses the `display_ticket()` method to show the Incident ID, Bot, and Description of each active incident ticket.

Search Incident:

Code Line:

```
for ticket in self.tickets:  
    if ticket.get_incident_id() == incident_id:
```

Explanation:

The program checks each ticket in the list and compares its Incident ID with the Incident ID entered by the user. If they match, the program displays the ticket that was found.

Remove Resolved Incident:

Code Line:

```
if ticket.get_incident_id() == incident_id:  
    self.tickets.remove(ticket)
```

Explanation:

The program searches for the Incident ID entered by the user. Once the matching ticket is found, the `remove()` method removes it from the list because the incident has already been resolved.

Count Active Incidents:

Code Line:

```
print("Total Active Incident Tickets:", len(self.tickets))
```

Explanation:

The `len()` function counts how many incident tickets are currently stored in the list. The program then displays the total number of active incident tickets.

III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.

START

>

Create IncidentTicketManager

>

Load Sample Tickets

>

Display Menu

>

Enter Choice

>

1. Add Ticket

```
>  
2. Display Tickets  
>  
3. Search Ticket  
>  
4. Remove Ticket  
>  
5. Count Tickets  
>  
6. Exit  
>  
END
```

IV. Conclusion: Briefly explain what you learned from the activity and how the linear data structure was applied.

This laboratory exercise helped me understand how a simple program can be used to manage incident tickets in an IT-related scenario. Through this activity, I was able to apply classes, objects, encapsulation, and methods while creating a program that can add, display, search, remove, and count incident tickets.

The linear data structure was applied through the Python list, where the incident ticket objects are stored in a sequence. By using the list, I was able to organize the tickets and perform the required operations on them.